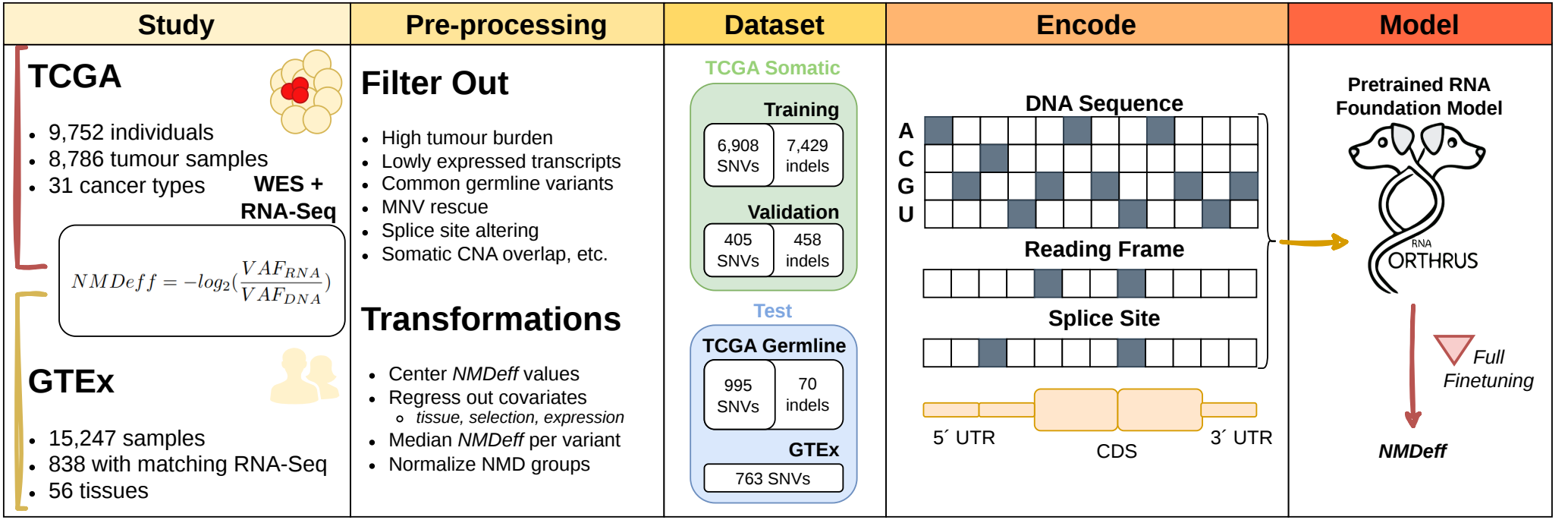
Study
Pre-processing
Dataset
Encode
Model
TCGA
9,752 individuals
8,786 tumour samples
31 cancer types
WES + RNA-Seq
$NMDeff = -log_2(\frac{VAF_{RNA}}{VAF_{DNA}})$
GTEx
15,247 samples
838 with matching RNA-Seq
56 tissues
Filter Out
High tumour burden
Lowly expressed transcripts
Common germline variants
MNV rescue
Splice site altering
Somatic CNA overlap, etc.
Transformations
Center NMDeff values
Regress out covariates
tissue, selection, expression
Median NMDeff per variant
Normalize NMD groups
TCGA Somatic
Training
6,908 SNVs
7,429 indels
Validation
405 SNVs
458 indels
Test
TCGA Germline
995 SNVs
70 indels
GTEx
763 SNVs
DNA Sequence
A
C
G
U
Reading Frame
Splice Site
5´ UTR
CDS
3´ UTR
Pretrained RNA Foundation Model
RNA ORTHRUS
Full Finetuning
NMDeff